

Computer Networks — One-Page-Per-Topic Cheat Sheet

Built for objective/MCQ recall (PSU & NMDC Steel ET pattern). Tables are the priority — memorise these, skim the prose.

1. OSI Model — Layers, Devices, Protocols, Units

#	Layer	Unit	Devices	Protocols / Function
7	Application	Data / Message	—	HTTP, FTP, SMTP, DNS, DHCP, SNMP, Telnet, SSH — user services
6	Presentation	Data	—	Translation, encryption, compression ; JPEG, MPEG, ASCII, SSL/TLS
5	Session	Data	—	Session establish/maintain/terminate, dialog control, synchronisation; RPC, NetBIOS
4	Transport	Segment	Gateway (loosely)	TCP, UDP, SCTP — end-to-end delivery, port addressing , segmentation, flow & error control
3	Network	Packet / Datagram	Router , L3 switch	IP, ICMP, IGMP, ARP*, RIP, OSPF, BGP — logical addressing & routing , fragmentation
2	Data Link	Frame	Switch , bridge, NIC	MAC, PPP, HDLC, Ethernet — framing, physical (MAC) addressing , error & flow control, access control
1	Physical	Bit	Hub , repeater, modem, cable	Bit transmission, encoding, topology, transmission mode, data rate

*ARP is commonly placed at the Data Link / Network boundary — most Indian PSU/GATE keys mark it **Network layer** (some books say Data Link). If options force a choice, go with Network layer.

Mnemonic (top→bottom): **All People Seem To Need Data Processing**.

Layer sub-divisions & key facts

- **Data Link = LLC + MAC** sublayers. **Router** works up to layer 3; **switch** layer 2; **hub** layer 1.
- Hub = broadcast, single collision domain, half duplex, dumb. Switch = MAC-table based forwarding, separate collision domain per port, full duplex.
- **Bridge** connects two LAN segments; **Router** connects different networks; **Gateway** connects dissimilar architectures (works at all layers).
- Reliable/connection-oriented service layers: Transport (TCP). Routing decisions: Network. Error detection in frames: Data Link.
- Encryption and compression → **Presentation** layer (frequently asked).
- Only layers present in both OSI and TCP/IP by same name: Transport, Network(Internet), Application.

OSI vs TCP/IP

Basis	OSI	TCP/IP
Layers	7	4 (or 5 in some texts)
Developed by	ISO (reference model)	DoD / ARPANET (practical, protocol-driven)
Merging	Separate Session, Presentation, Application	All three merged into Application
Physical+Data Link	Separate	Merged into Network Access / Link
Approach	Model defined first, then protocols	Protocols defined first, then model
Usage	Theoretical / teaching	Actually used on the Internet

2. Topologies, Media, Switching, Multiplexing

Topology	Links / Ports	Notes
Mesh	$n(n-1)/2$ links; $n-1$ ports each	Most reliable & robust, most expensive, most cabling
Star	n links, central hub	Easy to manage; single point of failure = hub
Bus	1 backbone + drop lines	Cheap, least cabling; hard fault isolation, needs terminators
Ring	n links, 2 ports each	Token passing; one break can kill the ring (dual ring fixes)
Tree / Hybrid	Hierarchical	Scalable; used in large networks

Transmission media

- **Guided**: Twisted pair (UTP/STP — cheapest, RJ45, Cat5/6), Coaxial (higher bandwidth, cable TV), **Optical fibre** (highest bandwidth, immune to EMI, no crosstalk, low attenuation, most secure, expensive; light propagation via **total internal reflection**; single-mode vs multi-mode).
- **Unguided**: Radio (omnidirectional), Microwave (line-of-sight, unidirectional), Infrared (short range, cannot pass walls — used in remotes).
- Attenuation measured in **dB**. Fibre uses light; coax/twisted pair use electrical signals.

Switching & multiplexing

Type	Key point
Circuit switching	Dedicated path reserved before transfer (telephone); 3 phases: setup, transfer, teardown. No store-and-forward. Fixed bandwidth, wasteful if idle.
Message switching	Whole message stored and forwarded hop-by-hop.
Packet - Datagram	Each packet routed independently, may arrive out of order (the Internet/IP).
Packet - Virtual circuit	Path fixed by setup, all packets follow it, in-order delivery (ATM, Frame Relay).
FDM	Frequency divided; analog; guard bands prevent overlap.
TDM	Time slots. Synchronous = fixed slots (wasted if idle). Statistical/Asynchronous = slots on demand, more efficient.
WDM	Multiple wavelengths of light on one fibre (FDM for optical).
CDM / CDMA	Unique codes, all users share full band simultaneously.

3. Data Link Layer — Error Control, Flow Control, Access

Error detection & correction

- **Parity (VRC)** — 1 bit; detects odd number of bit errors only; cannot detect even-numbered errors. **LRC** = block/2D parity.
- **Checksum** — 1's complement addition of segments, complement the sum; receiver adds all + checksum, result all 1s → no error. Used by **IP, TCP, UDP** (software-level).
- **CRC** — strongest detection method; based on **binary/modulo-2 division** by a generator polynomial. For a degree-n generator, append n zeros to data; remainder (n bits) is the CRC; receiver divides again — **remainder 0 = no error**. Generator of degree n → CRC is n bits. Detection only, no correction.
- **Hamming code** — detects and **corrects single-bit** errors. Redundant bits: $2^r \geq m + r + 1$. Parity bits placed at positions 1, 2, 4, 8, 16... (powers of 2). Hamming distance d → detects d-1 errors, corrects $\lfloor (d-1)/2 \rfloor$.

Flow control protocols — the comparison table

Protocol	Sender window	Receiver window	Notes
Stop-and-Wait	1	1	Simplest; very inefficient on long links
Go-Back-N (GBN)	$2^m - 1$	1	Cumulative ACK; on error retransmits whole window ; no out-of-order buffering
Selective Repeat (SR)	2^{m-1}	2^{m-1}	Independent ACKs; retransmits only the lost frame ; needs sorting/buffering; more complex

Formulas: $a = T_p / T_t$ (propagation ÷ transmission time).

Stop-and-Wait efficiency = $1 / (1 + 2a)$ · Sliding window efficiency = $N / (1 + 2a)$ (capped at 1), where N = sender window.

T_t = frame size / bandwidth · T_p = distance / propagation speed.

Multiple access / MAC protocols

Protocol	Key fact
Pure ALOHA	Transmit anytime; max efficiency 18.4% (1/2e); vulnerable time = $2 \times T_t$
Slotted ALOHA	Transmit only at slot start; max efficiency 36.8% (1/e); vulnerable time = T_t
CSMA	Sense before transmit. Variants: 1-persistent, non-persistent, p-persistent
CSMA/CD	Wired Ethernet (802.3); detects collision and aborts; needs min frame length; binary exponential backoff
CSMA/CA	Wireless (802.11); collision avoidance since detection is hard; uses RTS/CTS, ACK, IFS
Token Passing	Controlled access, collision-free, token circulates (Token Ring 802.5)
Polling / Reservation	Controlled access; primary station polls secondaries

- **MAC address:** 48 bits (6 bytes), hexadecimal, burned into NIC, **physical/flat** addressing. Broadcast MAC = FF:FF:FF:FF:FF:FF.
- Ethernet frame: min **64 bytes**, max 1518 bytes; payload (MTU) max **1500 bytes**.
- IEEE standards: **802.3** Ethernet, **802.4** Token Bus, **802.5** Token Ring, **802.11** WiFi/WLAN, **802.15** WPAN/Bluetooth, **802.16** WiMAX.

4. IP Addressing & Subnetting

IPv4 classes

Class	Range (1st octet)	Default mask	Networks	Hosts/network	Use
A	1 - 126	255.0.0.0 (/8)	$2^7 - 2$	$2^{24} - 2 = 16,777,214$	Very large orgs
B	128 - 191	255.255.0.0 (/16)	2^{14}	$2^{16} - 2 = 65,534$	Medium orgs
C	192 - 223	255.255.255.0 (/24)	2^{21}	$2^8 - 2 = 254$	Small networks
D	224 - 239	—	—	—	Multicast
E	240 - 255	—	—	—	Reserved / research

127.x.x.x is loopback — that is why Class A stops at 126. Identify class from the first 1-4 bits: 0=A, 10=B, 110=C, 1110=D, 1111=E.

Special & reserved addresses

- **Private:** 10.0.0.0/8 · 172.16.0.0-172.31.255.255 (/12) · 192.168.0.0/16
- **Loopback** 127.0.0.1 · **APIPA** 169.254.x.x (assigned when DHCP fails) · **This host** 0.0.0.0 · **Limited broadcast** 255.255.255.255
- Host-bits all **0** = network address; all **1** = broadcast address. Hence the -2 in usable hosts.

Subnetting mechanics

Host bits $h \rightarrow$ **usable hosts** = $2^h - 2$ · Borrowed bits $n \rightarrow$ **subnets** = 2^n
 $/x \rightarrow h = 32 - x$ · **Block size (increment)** = $256 - (\text{mask octet value})$
Network address = IP **AND** mask · Broadcast = network address with all host bits set to 1

Worked example — 192.168.10.100/26: /26 $\rightarrow$ mask 255.255.255.192; host bits = $6 \rightarrow 2^6 - 2 = 62$ **usable hosts**; block size = $256 - 192 = 64$; subnets are .0, .64, .128, .192. So 100 falls in the **.64** subnet $\rightarrow$ network **192.168.10.64**, broadcast **192.168.10.127**, usable range **.65 to .126**. Number of subnets from a /24 = $2^2 = 4$.

CIDR quick table

Prefix	Mask	Hosts	Prefix	Mask	Hosts
/24	255.255.255.0	254	/28	255.255.255.240	14
/25	255.255.255.128	126	/29	255.255.255.248	6
/26	255.255.255.192	62	/30	255.255.255.252	2
/27	255.255.255.224	30	/31, /32	—	special

IPv4 vs IPv6

Basis	IPv4	IPv6
Size	32 bits (4 bytes)	128 bits (16 bytes)
Notation	Dotted decimal	Hexadecimal, colon-separated (8 groups)
Header	20-60 bytes, variable	40 bytes fixed
Checksum	Present	Removed
Broadcast	Supported	No broadcast — uses multicast & anycast
Fragmentation	By sender and routers	Only by sender
Config / security	Manual or DHCP; IPsec optional	Auto-configuration (SLAAC); IPsec built in

IPv4 header — fields worth remembering

- Min **20 bytes**, max 60. **HLEN** in 4-byte words (so value 5 = 20 bytes).
- **TTL** — decremented by each router; at 0 the packet is discarded (prevents infinite loops). **Protocol** field identifies the upper layer (TCP=6, UDP=17, ICMP=1).
- Fragmentation fields: **Identification**, **Flags** (DF, MF), **Fragment Offset** (in units of 8 bytes).
- Header checksum covers the **header only**, not the data, and is recomputed at each router.

5. Network Layer Protocols & Routing

Protocol	Function
ARP	IP → MAC. Broadcast request, unicast reply. Maintains ARP cache
RARP	MAC → IP (obsolete, replaced by DHCP/BOOTP)
ICMP	Error reporting & diagnostics; used by ping (echo request/reply) and traceroute . Messages: destination unreachable, time exceeded, redirect, source quench
IGMP	Multicast group management
DHCP	Dynamic IP assignment; DORA = Discover, Offer, Request, Ack; ports 67 (server) / 68 (client)
NAT	Private ↔ public IP translation; conserves IPv4 addresses; done at the router

Routing protocols — know the algorithm for each

Protocol	Type	Algorithm	Key facts
RIP	Interior, Distance Vector	Bellman-Ford	Metric = hop count , max 15 hops (16 = infinity/unreachable); updates every 30 s; slow convergence, count-to-infinity problem
OSPF	Interior, Link State	Dijkstra	Metric = cost/bandwidth; each router keeps full topology map; areas; fast convergence; open standard
BGP	Exterior (between AS)	Path vector	The Internet's backbone routing protocol; policy-based
IGRP/EIGRP	Interior, hybrid	DUAL	Cisco proprietary

- **Distance vector**: knows only neighbours + distances, shares its whole table with neighbours. **Link state**: floods link info, each node builds the full map.
DV = "tell your neighbours about the world"; LS = "tell the world about your neighbours".
- **Static** routing = manual, no overhead, no adaptability. **Dynamic** = automatic, adapts to failures.
- Solutions to count-to-infinity: split horizon, route poisoning, hold-down timers.
- **Flooding** sends on all links except incoming. **Unicast / Multicast / Broadcast / Anycast** — know the definitions.

6. Transport Layer

Basis	TCP	UDP
Connection	Connection-oriented (handshake)	Connectionless
Reliability	Reliable — ACKs, retransmission	Unreliable, best-effort
Ordering	In-order guaranteed (sequence numbers)	No ordering
Header size	20-60 bytes (min 20)	8 bytes fixed
Flow / congestion control	Yes (sliding window, slow start)	None
Speed / overhead	Slower, high overhead	Faster , low overhead
Transmission	Byte stream	Message/datagram oriented
Uses	HTTP, FTP, SMTP, Telnet, SSH	DNS, DHCP, TFTP, SNMP, RIP, VoIP, streaming, online games

TCP essentials

- **Three-way handshake**: SYN → SYN+ACK → ACK. Connection release is normally **four-way** (FIN, ACK, FIN, ACK).
- Header fields: source/dest port, **sequence number**, acknowledgement number, HLEN, **flags** (URG, ACK, PSH, RST, SYN, FIN), window size, checksum, urgent pointer, options.
- **Congestion control**: slow start (exponential growth of cwnd) → congestion avoidance (additive/linear) → on loss, multiplicative decrease. Known as **AIMD**. Also fast retransmit & fast recovery.
- **Silly window syndrome** — fixed by Nagle's algorithm (sender side) and Clark's solution / delayed ACK (receiver side).
- **Socket** = IP address + port number. **Port** is 16 bits → 0 to 65535.
- Port ranges: **0-1023 well-known**, 1024-49151 registered, 49152-65535 dynamic/private.

7. Application Layer & Port Numbers

Port	Service	Transport	Port	Service	Transport
20 / 21	FTP (data / control)	TCP	80	HTTP	TCP
22	SSH / SFTP	TCP	110	POP3	TCP
23	Telnet	TCP	143	IMAP	TCP
25	SMTP	TCP	161	SNMP	UDP
53	DNS	UDP (TCP for zone transfer)	443	HTTPS	TCP
67 / 68	DHCP (server/client)	UDP	69	TFTP	UDP

Protocol notes

- **DNS** — hostname to IP. Hierarchy: root → TLD → authoritative. **Recursive** (server does the work) vs **iterative** (client follows referrals). Records: A (IPv4), AAAA (IPv6), MX (mail), CNAME (alias), NS, PTR (reverse), SOA.
- **HTTP** — **stateless**; methods GET, POST, PUT, DELETE, HEAD, PATCH, OPTIONS. State maintained by **cookies**. HTTP/1.1 added persistent connections.
- Status codes: **1xx** informational, **2xx** success (200 OK, 201 Created), **3xx** redirection (301, 304), **4xx** client error (**400** Bad Request, **401** Unauthorized, **403** Forbidden, **404** Not Found), **5xx** server error (500, 502, 503).
- **FTP** uses **two connections** (21 control, 20 data) — classic MCQ. Not secure; SFTP/FTPS are.
- **Email**: SMTP **sends/pushes**; POP3 and IMAP **retrieve/pull**. POP3 downloads and typically deletes from server; **IMAP keeps mail on the server** and syncs across devices. MIME allows non-ASCII attachments.
- **Telnet vs SSH**: Telnet is plaintext/insecure; SSH is encrypted. SSH is the correct answer for secure remote login.

8. Security & Miscellaneous

- **SSL/TLS** — operates between Application and Transport (often tagged Presentation layer in OSI questions); provides confidentiality, integrity, authentication. HTTPS = HTTP over TLS, port 443.

- **Firewall** — filters traffic by rules. Types: packet filtering (layer 3/4), stateful inspection, application/proxy gateway. **DMZ** = semi-trusted zone.
- **VPN** — secure tunnel over a public network; uses IPsec, tunnelling, encryption.
- **Attacks**: DoS/DDoS, MITM, IP spoofing, sniffing, phishing, replay, SYN flood.
- **Encryption**: symmetric (one shared key — DES, AES; fast) vs asymmetric (public/private key pair — RSA; slower). Digital signature uses the sender's **private** key. Hashing: MD5 (128-bit), SHA-1 (160-bit), SHA-256.

Performance terms

- **Bandwidth** = capacity (bps or Hz). **Throughput** = actual achieved rate (always $\leq$ bandwidth). **Latency** = total delay = propagation + transmission + queuing + processing delay. **Jitter** = variation in delay.
- **Bit rate** (bits/sec) vs **Baud rate** (signal changes/sec): $\text{bit rate} = \text{baud rate} \times \text{bits per signal}$. Bit rate $\geq$ baud rate.
- **Nyquist** (noiseless): $C = 2B \log_2 L$. **Shannon** (noisy): $C = B \log_2(1 + \text{SNR})$.
- **Simplex** (one direction) · **Half duplex** (both, not simultaneously — walkie-talkie) · **Full duplex** (simultaneous — telephone).
- **Network types**: PAN < LAN < CAN < MAN < WAN. LAN = high speed, low error, privately owned; WAN = large area, leased lines, higher delay.
- **Client-server vs peer-to-peer**; **MTU** = largest frame payload (Ethernet 1500 bytes).

If you revise only five things from this sheet: (1) the layer-device-protocol-unit table, (2) port numbers, (3) IP classes + the subnetting formulas, (4) TCP vs UDP, (5) routing protocol → algorithm mapping. Those five carry the majority of CN marks in PSU papers.